

Odds and Ends

Last day of class is usually a mix of trying to find gaps or fill in holes.

I discussed phylogenetic tree building.

```
git clone https://github.com/biodataprogram/GEN220_2025_classexamples.git
```

```
cd GEN220_2025_classexamples/Trees
```

```
module load muscle
```

```
module load fasttree
```

```
module load iqtree
```

```
module load trimal
```

```
module load clipkit
```

```
# build an alignment of sequences already identified as homologs
```

```
# previously I had started with MET12 (S. cerevisiae) enzyme
```

```
more MET12.fa # single sequence
```

```
ls -l MET12.hit_seqs.fasta # the collection of homologs for MET12 in a few yeast fungi
```

```
# denovo multiple alignment - writes in multi-fasta format
```

```
muscle -align MET12.hit_seqs.fasta -output MET12.hit_seqs.fasaln
```

```
# trim sequences - using automated parameters - see http://trimal.cgenomics.org/trimal for more
```

```
trimal -automated1 -in MET12.hit_seqs.fasaln -out MET12.hit_seqs.mfa.trim
```

```
# this is an alternative alignment trimmer
```

```
clipkit MET12.hit_seqs.fasaln
```

```
# build a tree w fasttree (FastTreeMP uses multiple processors, FastTree uses 1 processor only)
```

```
FastTreeMP < MET12.hit_seqs.fasaln > MET12.hit_seqs.tre
```

```
# build a tree with IQ-TREE2 - ultrafast bootstrap and first determine optimal number of processors
```

```
iqtree3 -s MET12.hit_seqs.fasaln -nt 2 -bb 1000 -alrt 1000
```

Some links

- [Muscle](#) - Multiple alignment tool
- [TrimAl](#) - alignment trimming tool
- [clipkit](#) - alignment trimming tool
- [HMMER](#) - HMMER - Hidden Markov Model for biosequence analyses.
- [FastTree](#) - Fast Phylogenetic Tree construction
- [IQ-TREE](#) - Phylogenetic Tree construction
- [RAxML](#); [a tutorial](#)
- [iTOL](#) - Tree visualization (web-based) tool
- [FigTree](#) - Tree visualization (can run on HPCC if you have X11 enabled:
module load figtree; figtree)
- [ggtree](#) - R package for Tree rendering

```
module load hmmer
```

```
# build an HMM from a multiple alignment
```

```
hmmbuild MET12.hmm MET12.hit_seqs.fasaln
```

This is a little circular I am searching the HMM back against the original sequences, but if you wanted to instead search this HMM against a database of proteins (eg swissprot or your collection of proteins from species)

```
module load hmmer
module load samtools
```

```
# domtbl is the result file which has columns of data that are parseable instead of more complex
hmmsearch -E 1e-3 --domtblout MET12.search.domtbl MET12.hmm DATABASE > MET12.search.hmmsearch
```

```
# you could download a database like swissprot
```

```
#curl -O https://ftp.uniprot.org/pub/databases/uniprot/current_release/knowledgebase/complete
```

```
#gunzip uniprot_sprot.fasta.gz
```

```
DB=uniprot_sprot.fasta
```

```
# or one on the cluster
```

```
cp /srv/projects/db/Swissprot/2023_03/uniprot_sprot.fasta $DB
```

```
hmmsearch -E 1e-3 --domtblout MET12.search.domtbl MET12.hmm $DB > MET12.search.hmmsearch
```

```
# retrieve these hits
```

```
grep -v ^# MET12.search.domtbl | awk '{print $1}' | sort -u | samtools faidx -r - uniprot_sprot.fasta
```

```
# to align a set of proteins back to an HMM (which is instead of doing a denovo multiple alignment)
```

```
hmmalign MET12.hmm MET12.sprot_hits.fasta > MET12.sprot_hits.stk
```

```
# convert the stockholm format to multifasta (afa)
```

```
esl-reformat afa MET12.sprot_hits.stk > MET12.sprot_hits.fasaln
```

```
# convert the stockholm format to clustal
```

```
esl-reformat clustal MET12.sprot_hits.stk > MET12.sprot_hits.clustalw
```

```
# rebuild a tree
```

```
clipkit MET12.sprot_hits.fasaln
```

```
# build a tree w fasttree (FastTreeMP uses multiple processors, FastTree uses 1 processor on a single node)
```

```
FastTreeMP < MET12.sprot_hits.fasaln.clipkit > MET12.sprot_hits.tre
```

```
# read this tree
```

```
cat MET12.sprot_hits.tre
```

```
# try copy and pasting this and opening in iTOL
```

```
# https://itol.embl.de/
```